



C++ Programming

CERTIFICATION COURSE

**Master C++ Programming Skills and
Elevate Your Career to New Level !**

C++ Programming

The C++ Programming Course by W3 Technologies teaches you to write efficient, scalable code using both procedural and object-oriented techniques. You'll focus on C++ syntax, memory management, pointers, classes, and inheritance, while learning best practices for maintainable programs. The course equips you with the skills to create flexible and reusable code by building a solid understanding of object-oriented principles.

Why W3 Technologies

Training at W3 Technologies equips you to excel in today's competitive world. From the very beginning of your journey with us, you'll experience continuous growth through learning. We provide support throughout the course, while encouraging you to become independent and confident in your skills.



Lower Learning Cost

Learning with us is budget-friendly. Our courses are affordable yet provide comprehensive and engaging content.



Expert Instructors

Learning with us means benefiting from top-tier expertise. Our expert instructors provide deep knowledge and engaging lessons.



Certification

Earn an industry-recognized certification upon completing the course, enhancing your resume.

Introduction

- History, Features
- Rules of C++ programming
- Structure of C++ program
- C++ Tokens
- Identifiers, Keywords, Constants, Operators, Special characters
- C++ Data types

Basic programming

- Console I/O Statements(cin, cout)
- Programs to perform various calculations
- Operators
- Programs to implement various operators

Control statements

- Conditional Control Statements
- If-else , switch-case
- Loops
- While, do while, for
- Implementing programs on conditional & loops
- break, continue, goto keywords

Arrays

- Definition, advantages
- Array types
- Single dimension
- Double dimension
- Declaration, accessing array data
- Implementation of array operations

Working with Functions

- Definition, advantages, types of functions, classification
- Implementing various kinds of functions
- Inline functions

Function Overloading

- What is function over loading
- Implementing overloading on various functions

Operator Overloading

- Definition, why it is required, advantages
- About operator keyword, rules of operator overloading
- Overloading various operators

Constructors & Destructors

- Definition, uses
- Types (Default Constructor, Parameter Constructor, Copy Constructor)
- Destructors

Friend Function & Friend classes

- Friend Function definition, usage of friend keyword
- Implementing of friend functions in various scenarios
- Friend Class definition, usage
- Implementing of friend class

Inheritance

- Definition, Advantages
- Types of Inheritances (Single, Hierarchical, Multilevel, Multiple Hybrid)
- Implementing various kinds of inheritances
- Implementing various constructors in inheritance
- Implementing function overriding
- Implementing various access levels in inheritance

Virtual Functions & Abstract Classes

- Definition, requirement
- How to implement
- Pure virtual function definition
- Implementing abstract classes

Templates

- Template Definition
- Generic Function
- Generic Class
- Template function Overloading

I/O Streams & Files

- Streams Hierarchy
- Input Streams & Output Streams
- What is File?
- Implementing various file operations on basic data types(write, read, append and modify)
- Implementing various file operations on object data types(write, read, append and modify)
- Random Access Files(seekp, seekg, tellp, tellg)

Formatted IO

- IO Flags
- IO Functions

Exception Handling

- Try, Catch and Throw

Thank You for Choosing **W3 Technologies**

Thank you for choosing to explore this course from W3 Technologies. We hope you find it insightful and valuable as you embark on your learning journey.

At W3 Technologies, we believe that learning is a continuous process, and we are excited to be part of your journey towards gaining new skills and knowledge.



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